Bioinformatics algorithms

Rosario M. Piro
Politecnico di Milano
Genome assembly – exercises

Exercise 1.

Recall our definitions of our idealized genome from the lecture.

- G: genome size
- L: read length
- N: read number
- n_b: total number of total bases (analogous for k-mers)
- d_b: mean base depth (analogous for k-mers)

Let G = 5 Megabases (approximately the genome of *E.coli*), L = 200 bases

(Note: *Escherichia coli* is a bacterion which is frequently used as a "model organism", especially for basic biochemical functions and pathways.)

In the lecture, we calculated the sequencing depth needed such that at least 99% of the human genome would be covered by at least one read. Under the same slightly unrealistic assumptions as in the lecture, calculate the number of 200 bp reads required to ensure that at least 95% of the *E. coli* genome is covered.

Exercise 2.

Refering to the formulas from the lecture, explain why demanding a larger overlap between reads in order to combine reads into contigs is associated with a lower expected contig size.

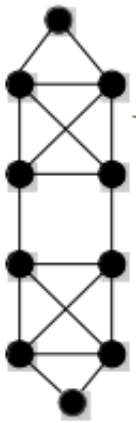
Exercise 3.

a) Construct a de Bruijn graph for the following short DNA sequence, using a decomposition of the sequence into 4-mers:

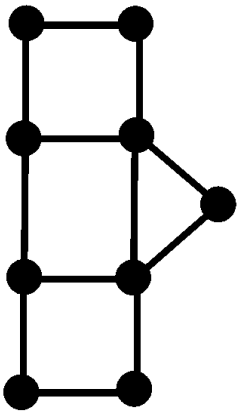
```
A C G T T A C G T A G G T T C A G T A T C
```

Exercise 4.

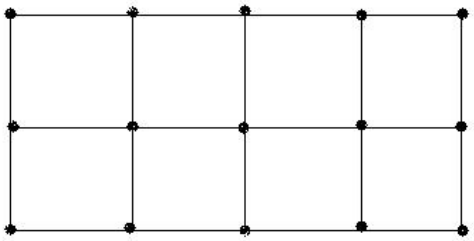
a) Does the following graph have a Eulerian path? Does it have a Eurlerian cycle?



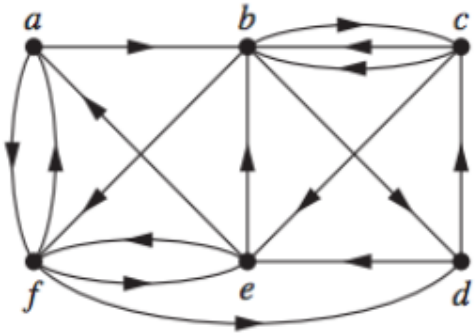
b) Does the following graph have a Eulerian path? Does it have a Eurlerian cycle?



c) Does the following graph have a Eulerian path? Does it have a Eurlerian cycle?



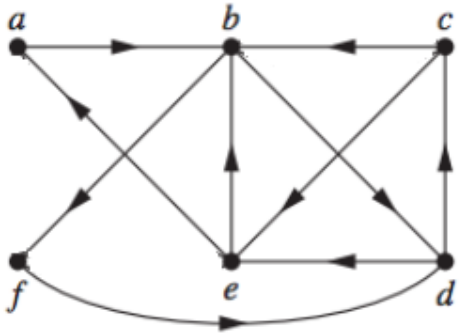
d) Does the following graph have a Eulerian path? Does it have a Eurlerian cycle?



e) Take one of the three undirected graphs (a-c) which has a Eulerian path but not a Eulerian cycle.

- Can you slightly modify the graph such that it has a Eulerian cycle? If so, how?
- What if the graph was a directed graph with a Eulerian path, could you still modify it to become a Eulerian cycle? If so, how?

Exercise 5.



For the directed graph shown above, use Hierholzer's approach to identify a Eulerian path!